

*Standard Operating Procedure **MIG 255 Welder***

Lincoln Power MIG 255C

The Power MIG 255c is an industrial-grade wire feed welder featuring **Diamond Core Technology**, providing a forgiving arc and excellent out-of-position welding. It delivers an output range of **30–300 Amps** and is designed for fabrication on mild steel, stainless steel, and aluminum

Potential Hazards

- **Electric Shock:** The welding circuit is electrically "hot" when the welder is on; contact can cause severe injury or death.
- **Arc Rays:** Ultraviolet and infrared rays can burn eyes and skin instantly (Arc Eye).
- **Fumes and Gases:** Welding produces hazardous fumes; concentrated inhalation can cause lung damage or "metal fume fever".
- **Fire/Explosion:** Sparks and spatter can travel up to **35 feet** and ignite flammable materials or explosive atmospheres.
- **Moving Parts:** The wire drive rolls and cooling fan present pinch and entanglement points

Personal Protective Equipment (PPE)

PPE Item	Requirement
Welding Helmet	**ANSI Z87.1 Compliant:** Must use a properly fitted helmet with the correct grade of filter plate (Shade 10-13 depending on amperage) to protect against arc rays.
Eye Protection	Wear ANSI Z87.1-compliant safety goggles. Mandatory under the helmet to protect against slag and debris when the helmet is raised.
Hand Protection	**Dry, Hole-Free Welding Gloves:** Leather TIG gloves are required to insulate hands from heat and "hot" electrical circuits.



*Standard Operating Procedure **MIG 255 Welder***

PPE Item	Requirement
Body Protection	**Flame-Resistant Apparel:** Wool or heavy cotton clothing, leather aprons, and high boots are required to prevent burns from arc flash and spatter.
Respiratory	**Fume Extraction: Use local exhaust ventilation to keep the breathing zone clear of welding fumes.**

Before Use

- **Workspace Inspection:** Ensure the area is dry and free of flammable materials, tools, and unnecessary items.
- **Machine Setup:** Check the drive roll tension and ensure the wire is feeding smoothly through the liner.
- **Polarity Check:** Verify the machine is set to **DC+** (Electrode Positive) for MIG or **DC-** for most Flux-Cored wires.
- **Gas Calibration:** Open the cylinder valve slowly; set the flow meter between **25–35 CFH** (Cubic Feet per Hour) for Argon/CO2 mixes.
- **Work Connection:** Ensure the work clamp is attached to clean, paint-free metal as close to the weld area as possible.
- **Condition Report:** Report any frayed cables, cracked torch handles, or gas leaks to the shop supervisor immediately.

During Use

- **Focus:** Concentrate on the weld puddle and arc; avoid distractions while the trigger is engaged.
- **Settings Adjustment:** Refer to the chart inside the wire drive door for correct Voltage (V) and Wire Feed Speed (WFS) based on material thickness.
- **The "4-Inch" Rule:** Keep your free hand at least **4 inches** away from the contact tip and the arc at all times.



*Standard Operating Procedure **MIG 255 Welder***

- **Nozzle Distance:** Maintain a 1/4 to 3/8 inch stick-out (distance from the contact tip to the workpiece) for stable arc performance.
- **Thermal Duty Cycle:** Be mindful of the machine's duty cycle (40% at 250A); if the thermal overload light illuminates, allow the fan to run while the machine idles.
- **Cylinder Safety:** Never touch the electrode or welding wire to the gas cylinder.

After Use

- **System Shutdown**: Turn off the power switch and close the gas cylinder valve.
- **Line Purge**: Depress the foot pedal briefly to purge remaining gas from the lines before storage.
- **Housekeeping**: Coil cables neatly on the side-mounted torch holders. Sweep the area of metal off-cuts and debris.
- **Hygiene**: Wash hands thoroughly to remove metallic oxides and particulates after handling filler rods and workpieces.

Emergency Equipment and Procedures

- **Emergency Stop**: Immediately release the foot pedal or trigger and toggle the main power switch to the **OFF** position.
- **Thermal Overload**: If the **Thermal Shutdown Light** illuminates, allow the fan to run with the machine idle until it cools; do not shut off power immediately as this stops the cooling fan.
- **Reporting**: Report all electrical shocks, near-misses, burns, or equipment damage to the Laboratory Project Supervisor.